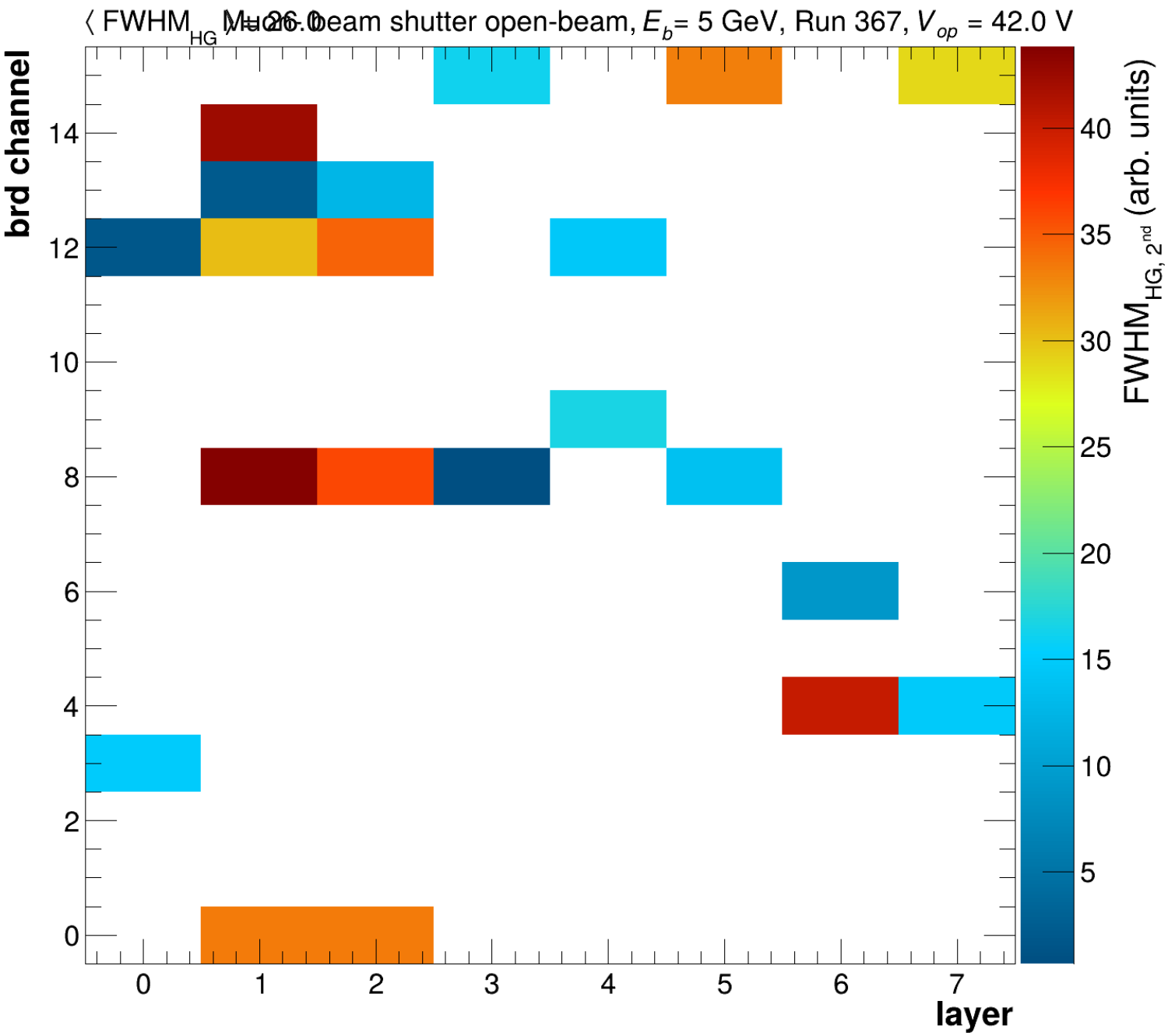
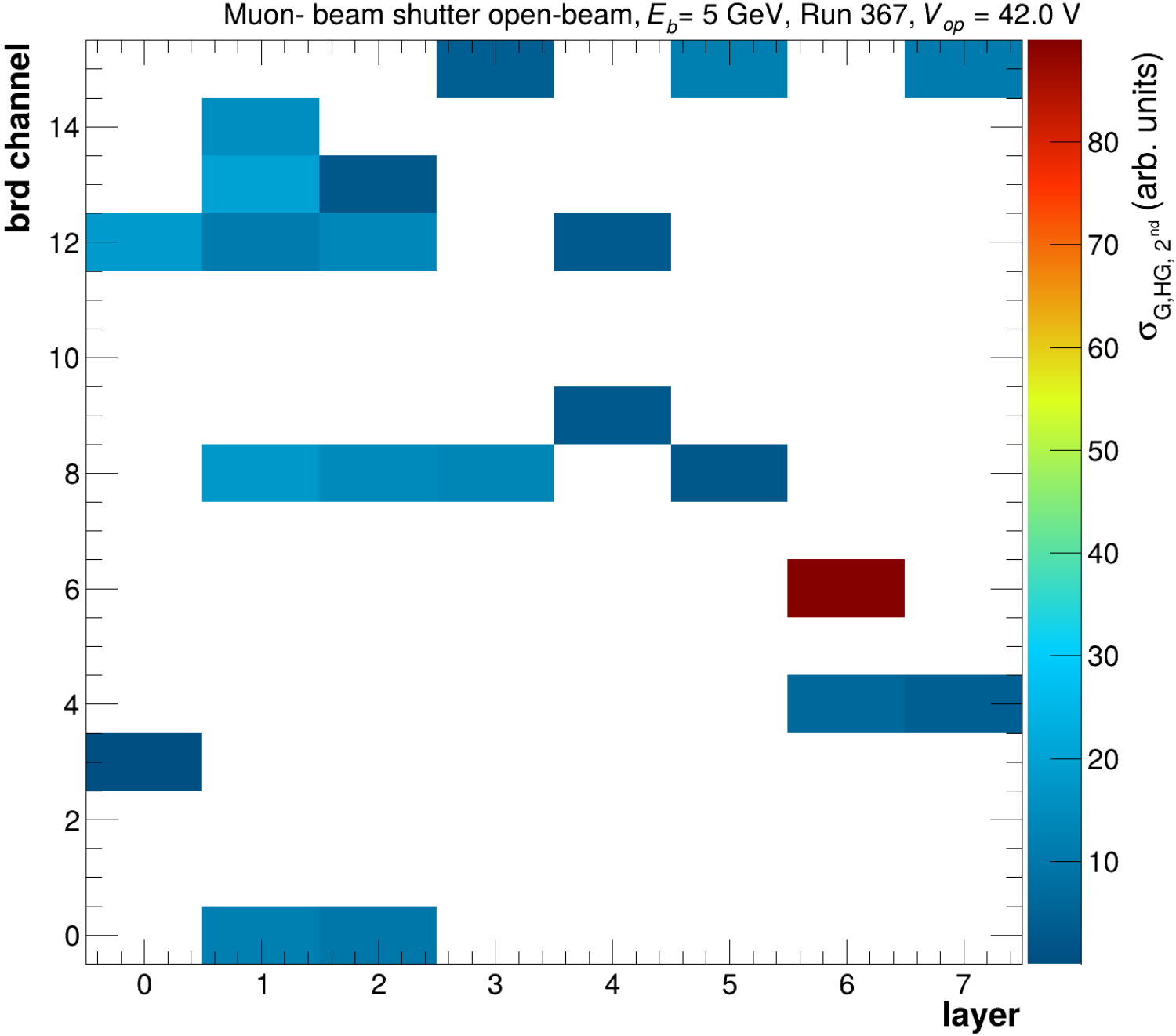
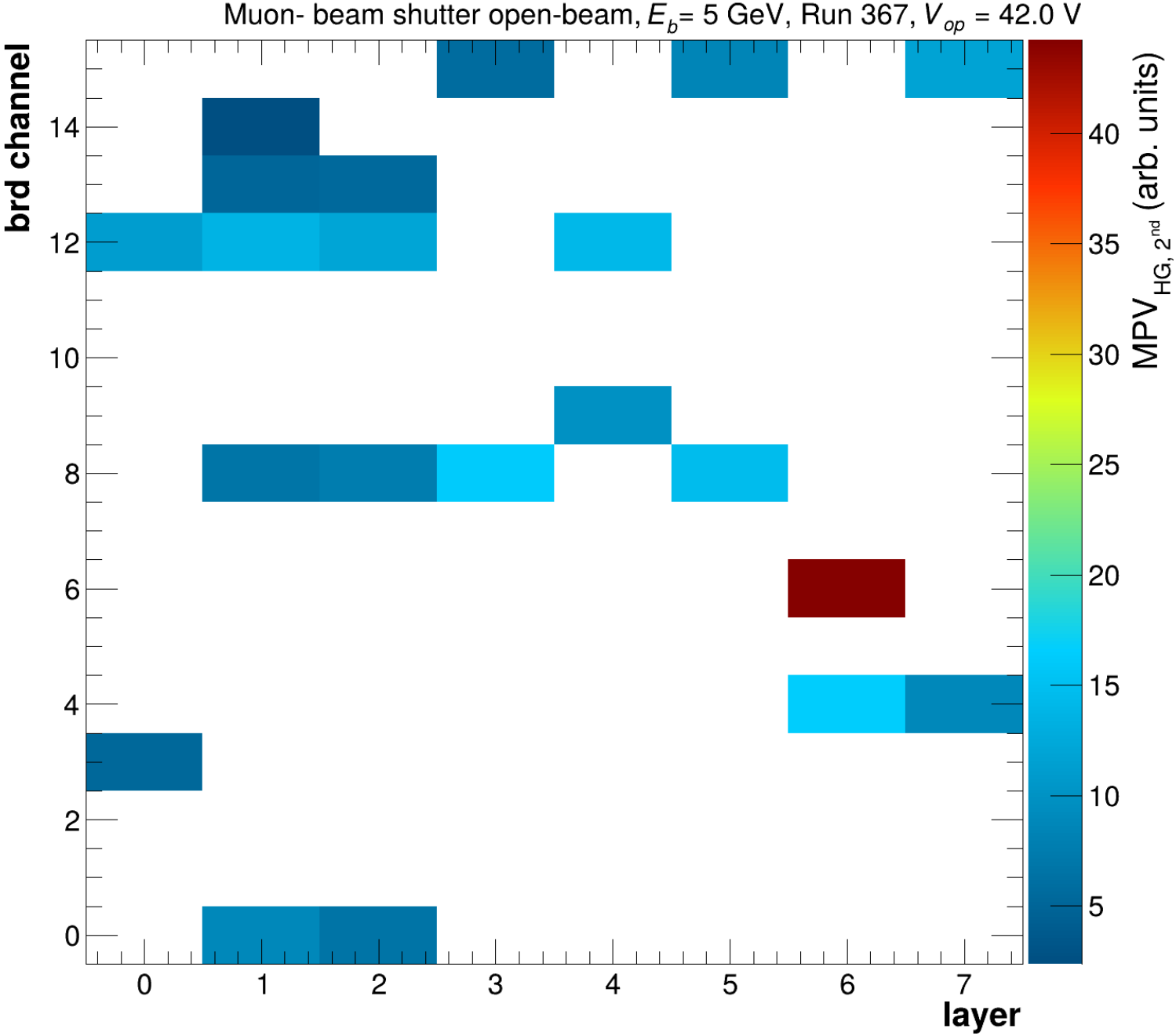
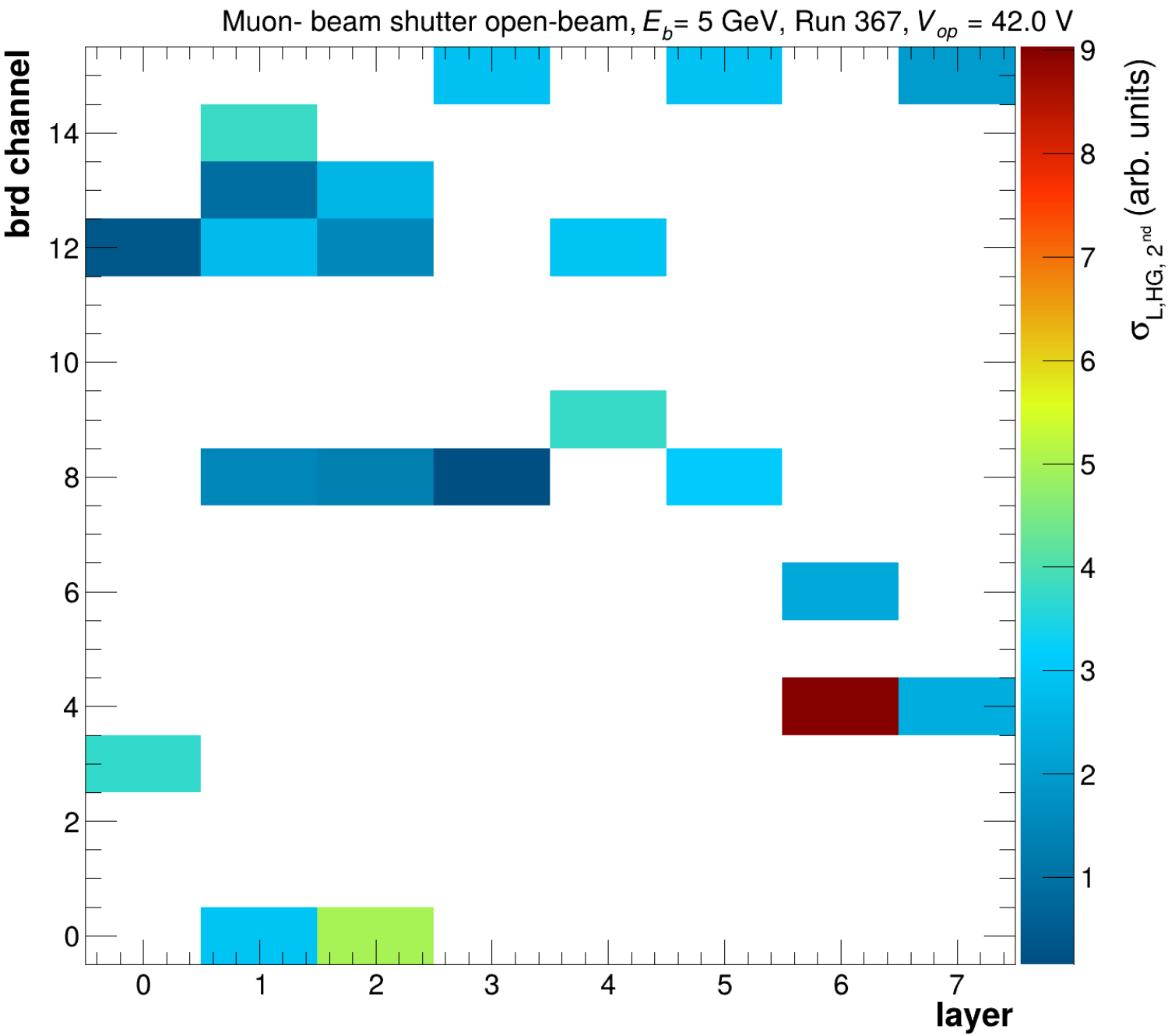
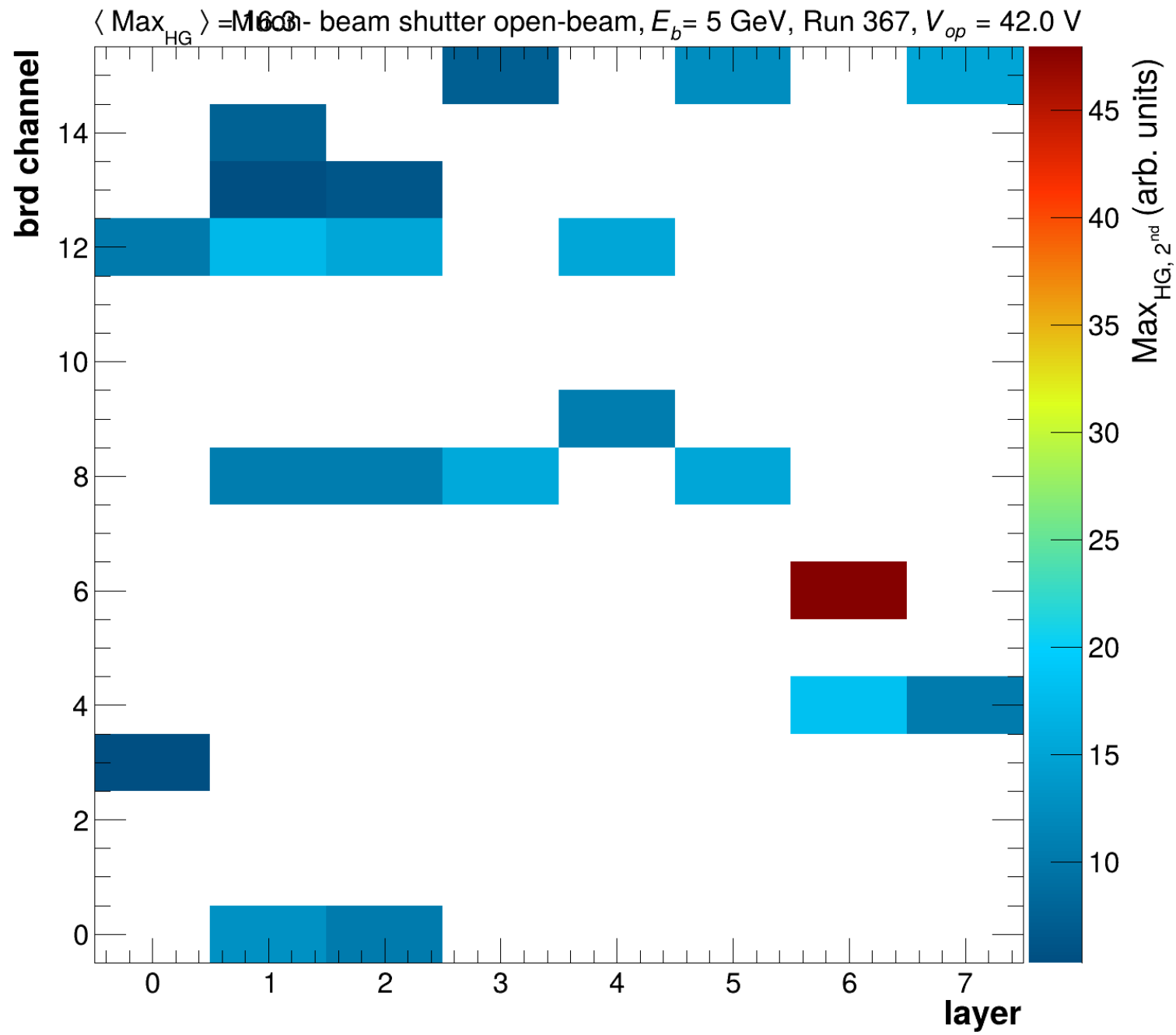
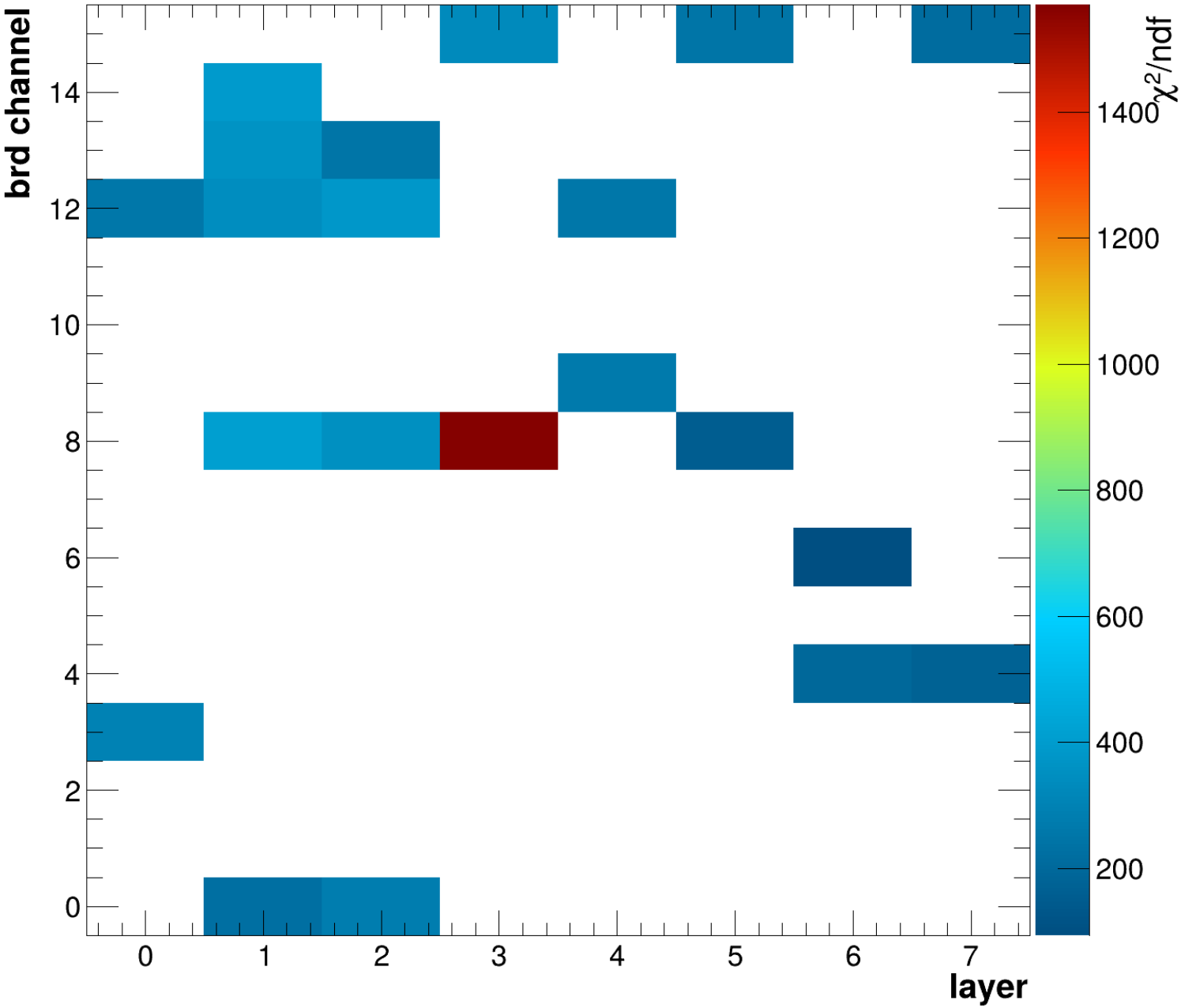

⟨ FWHM_HG ⟩ = 26.0
beam shutter open-beam, E_b = 5 GeV, Run 367, V_op = 42.0 V
brd channel
layer
FWHM_HG, 2nd (arb. units)
0
1
2
3
4
5
6
7
0
2
4
6
8
10
12
14
5
10
15
20
25
30
35
40

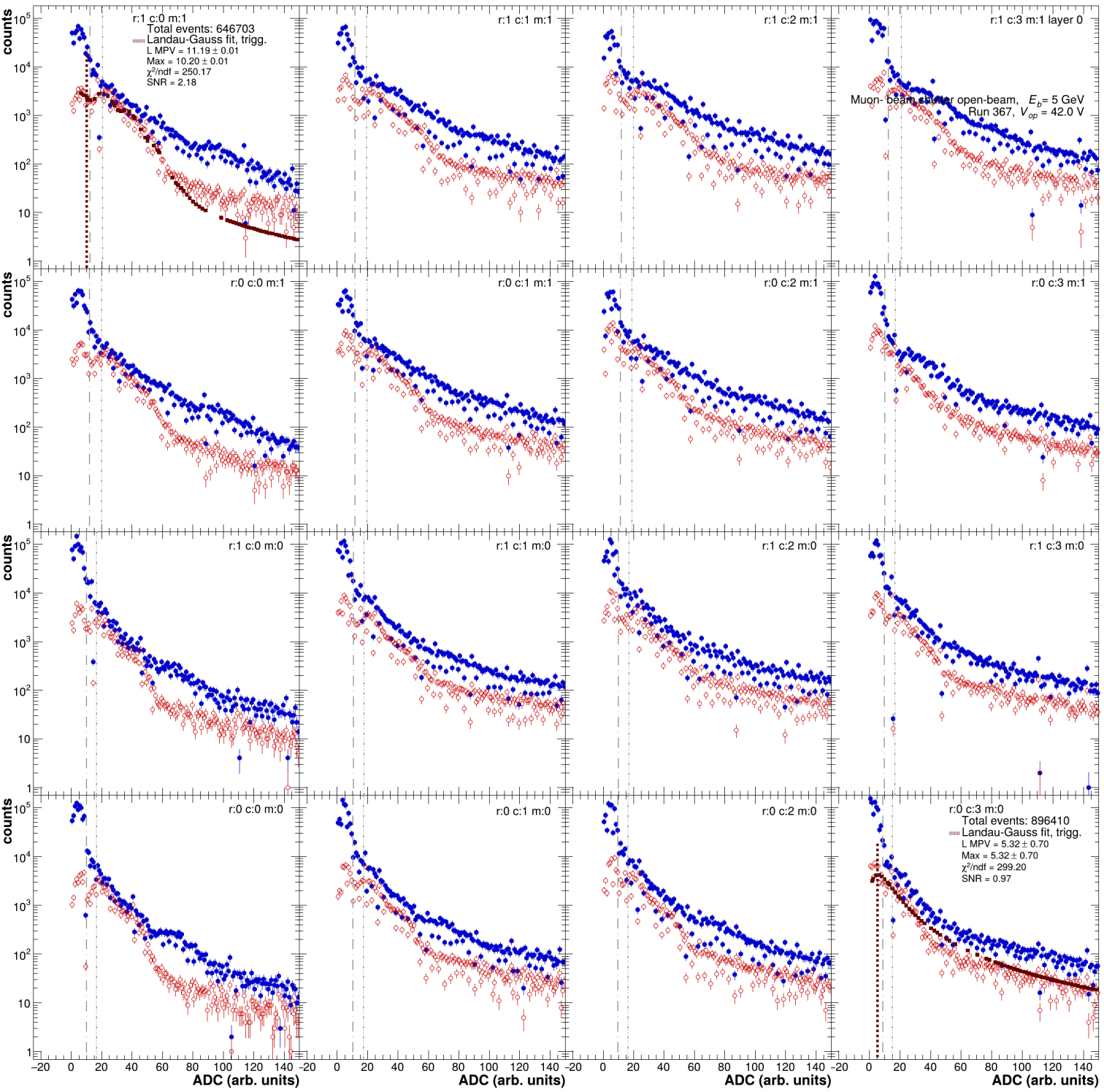

Muon- beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
layer
$\sigma_{G,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
20
30
40
50
60
70
80

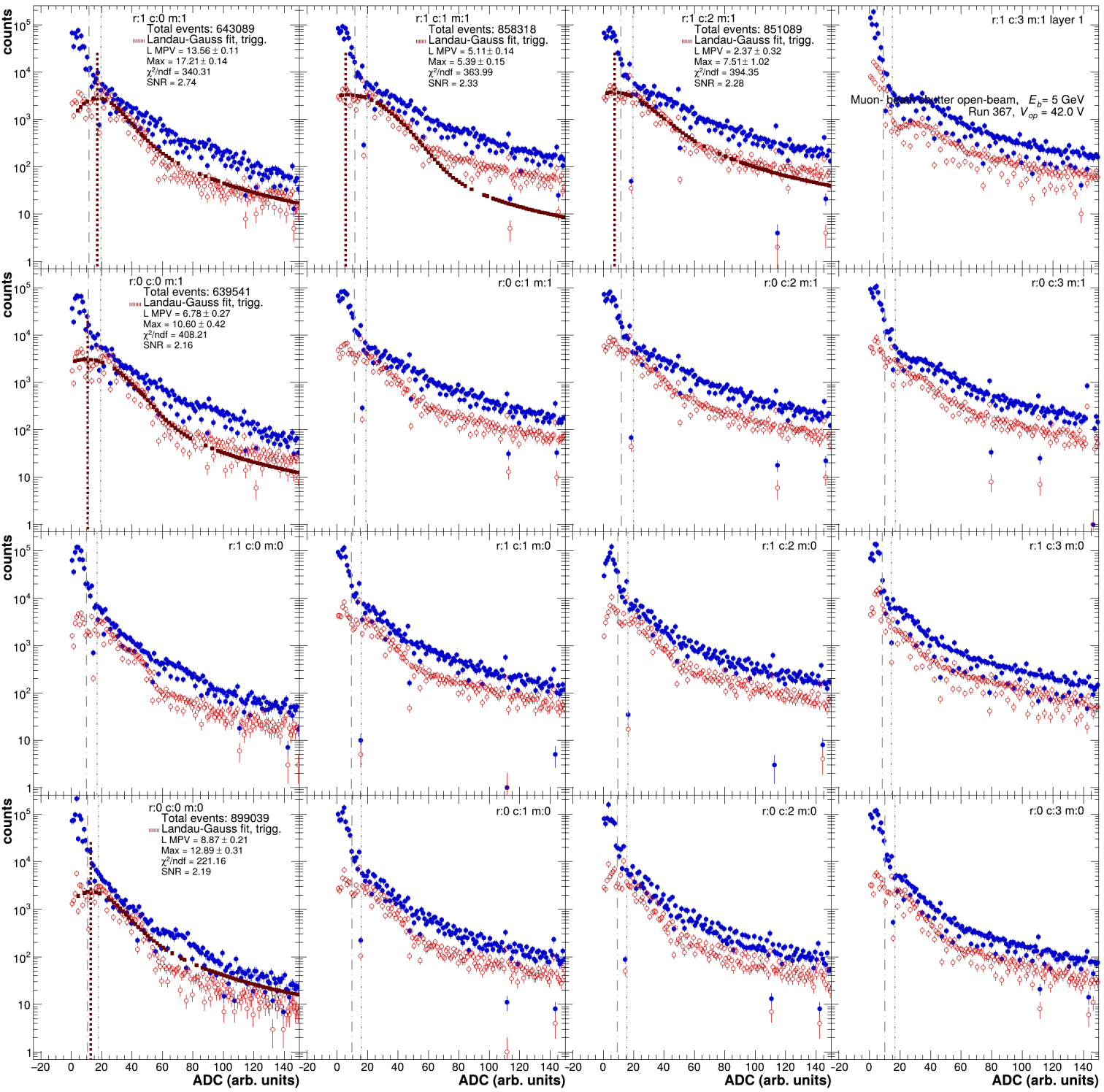

Muon- beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
layer
$MPV_{HG,\ 2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
5
10
15
20
25
30
35
40

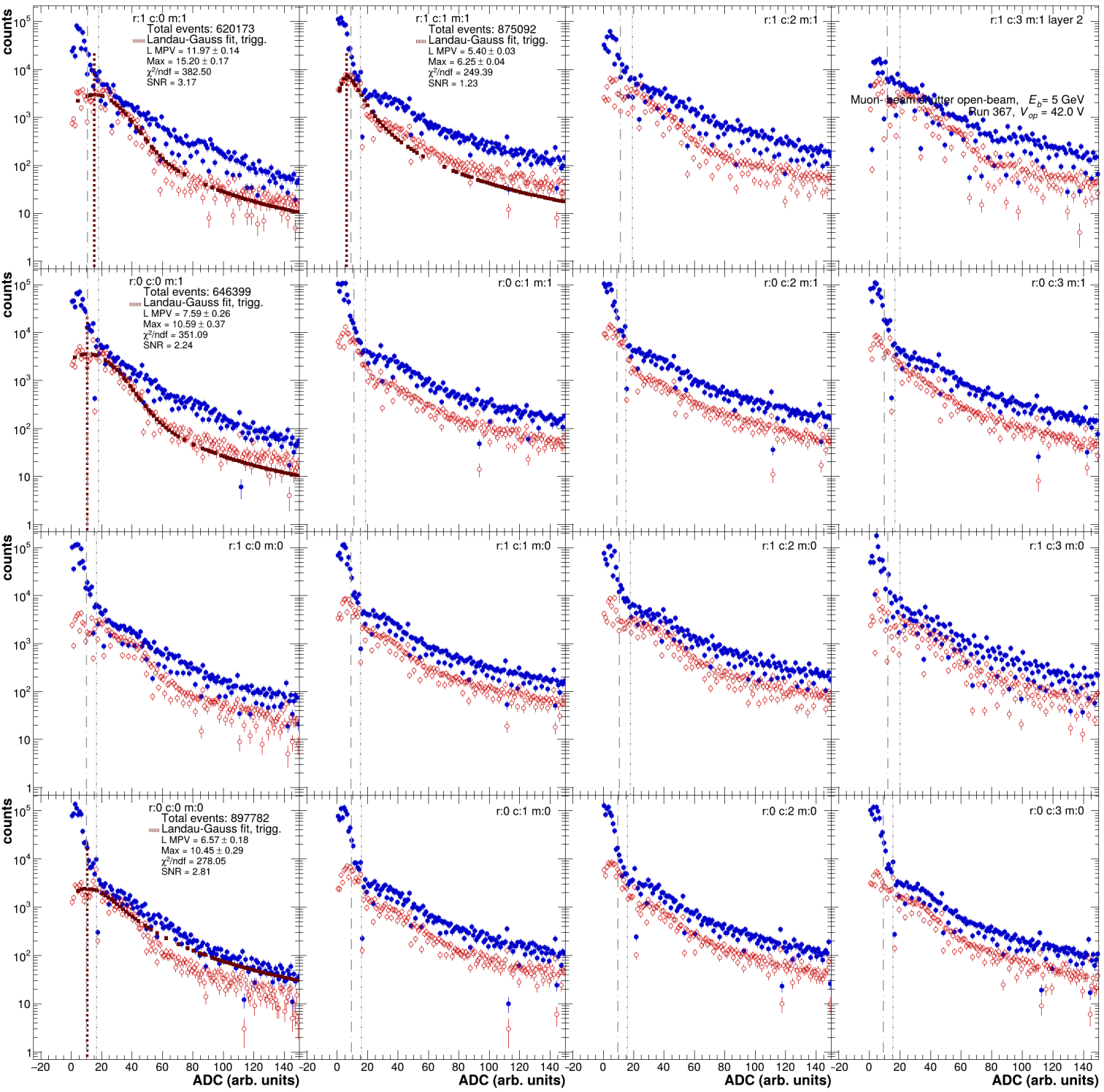

Muon- beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
layer
$\sigma_{L,HG,2^{nd}}$ (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
8
9

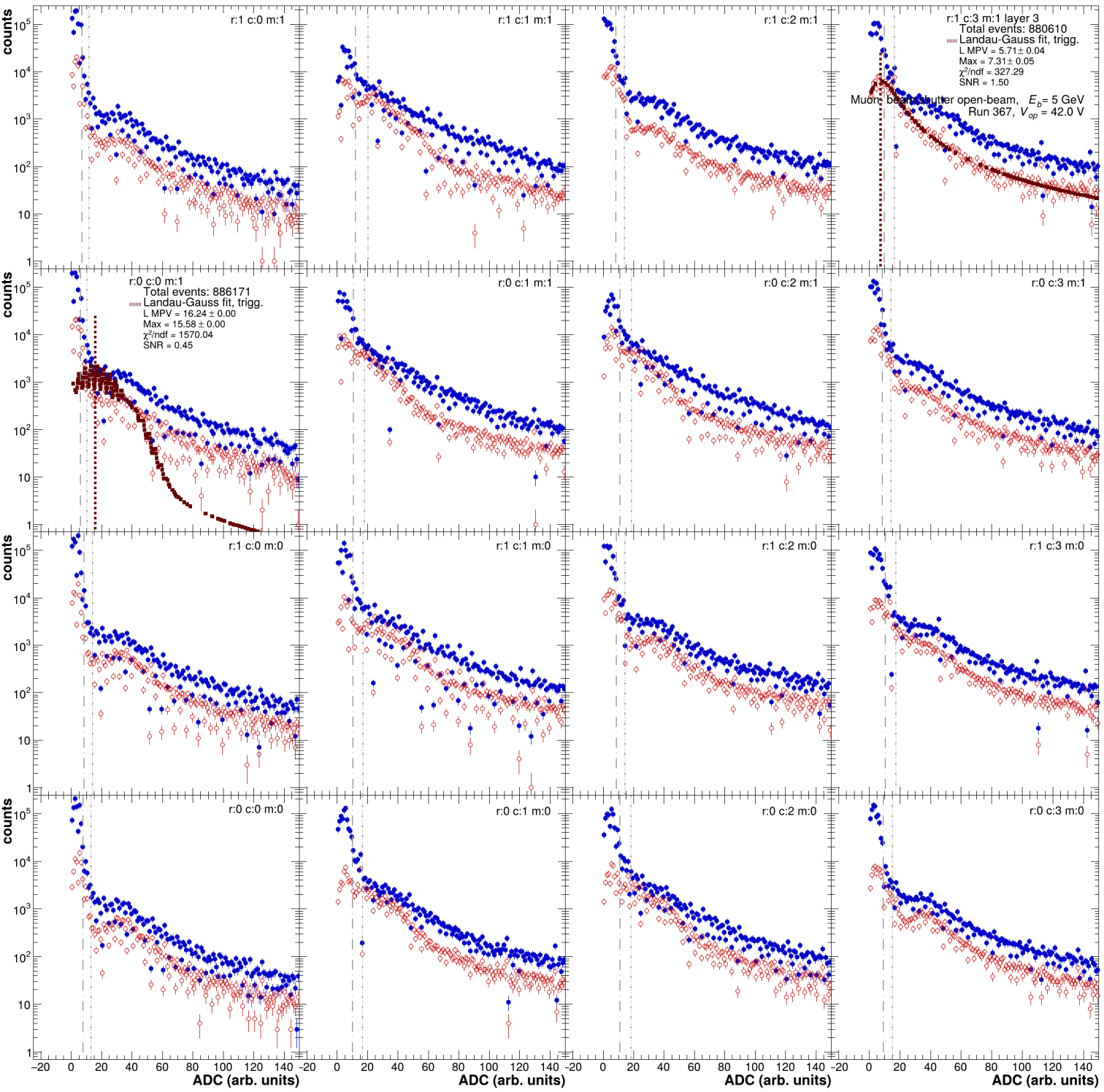

⟨ Max_HG ⟩ = 16.3 - beam shutter open-beam, E_b = 5 GeV, Run 367, V_op = 42.0 V
brd channel
layer
Max_HG, 2nd (arb. units)
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
10
15
20
25
30
35
40
45

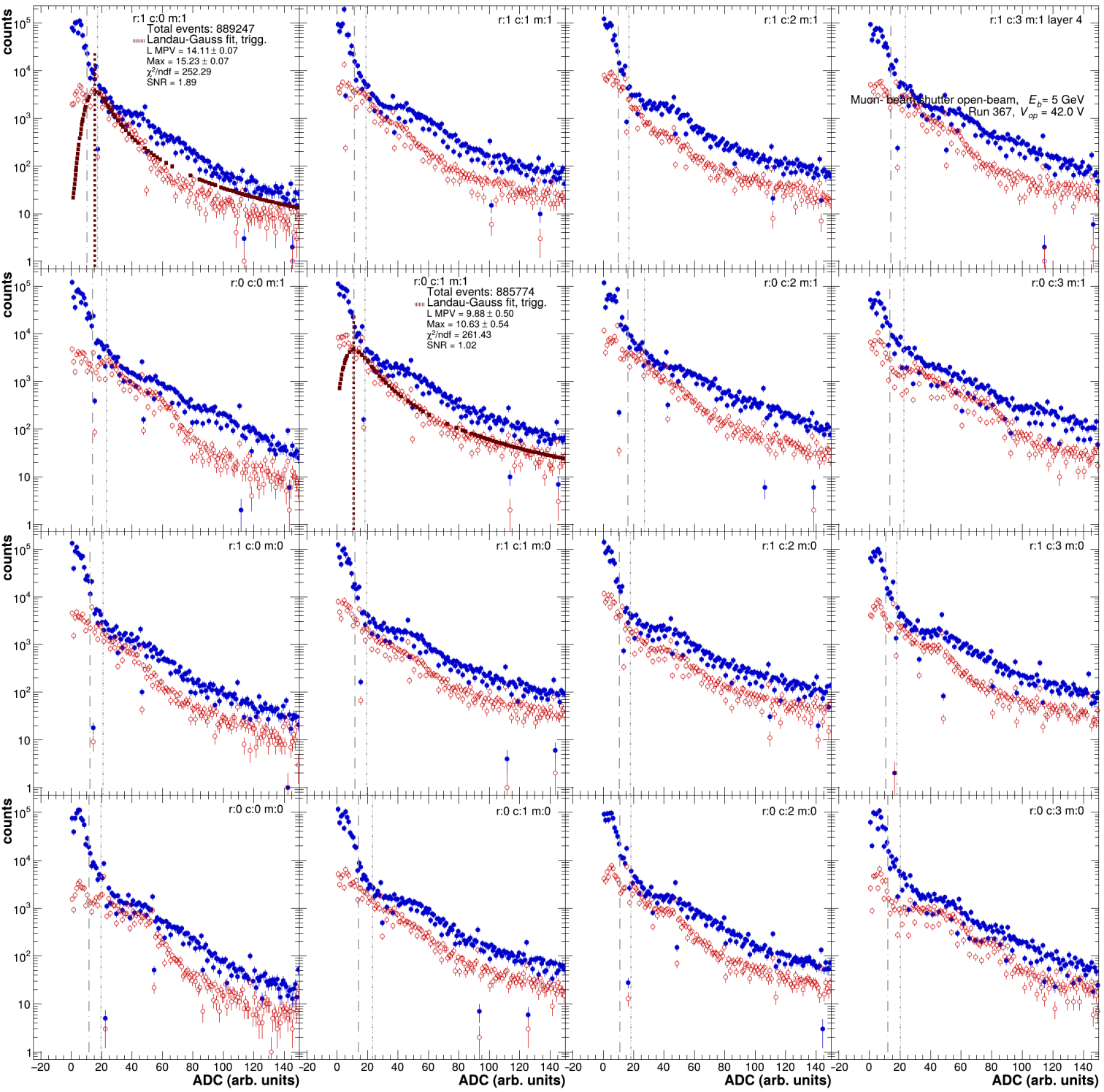

Muon- beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
layer
χ^2/ndf
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
200
400
600
800
1000
1200
1400

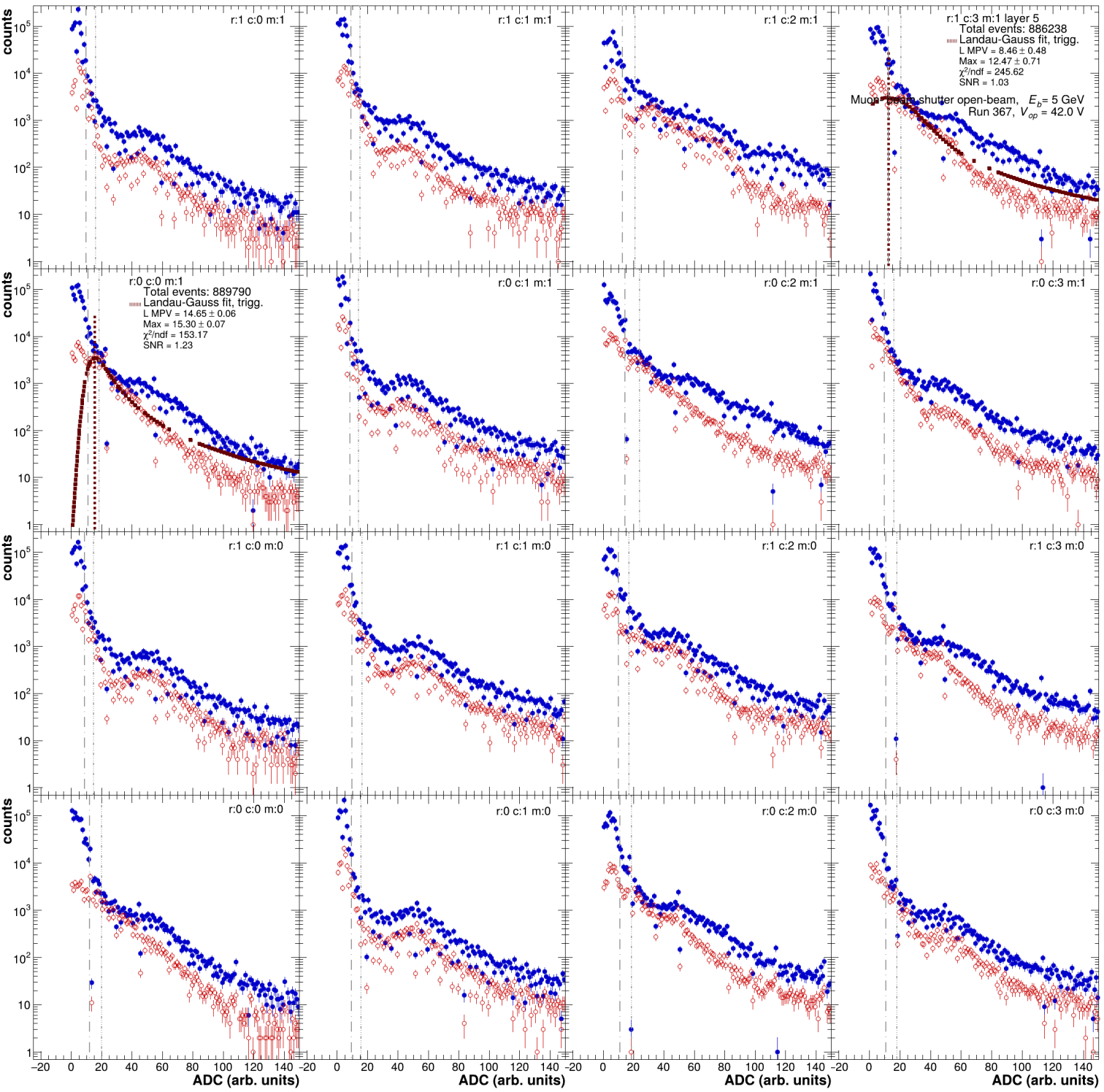

r:1 c:0 m:1
Total events: 646703
Landau-Gauss fit, trigg.
L MPV = 11.19 ± 0.01
Max = 10.20 ± 0.01
χ²/ndf = 250.17
SNR = 2.18
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
Muon- beam, shutter open-beam, E_b = 5 GeV
Run 367, V_{op} = 42.0 V
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
Total events: 896410
Landau-Gauss fit, trigg.
L MPV = 5.32 ± 0.70
Max = 5.32 ± 0.70
χ²/ndf = 299.20
SNR = 0.97
counts
ADC (arb. units)

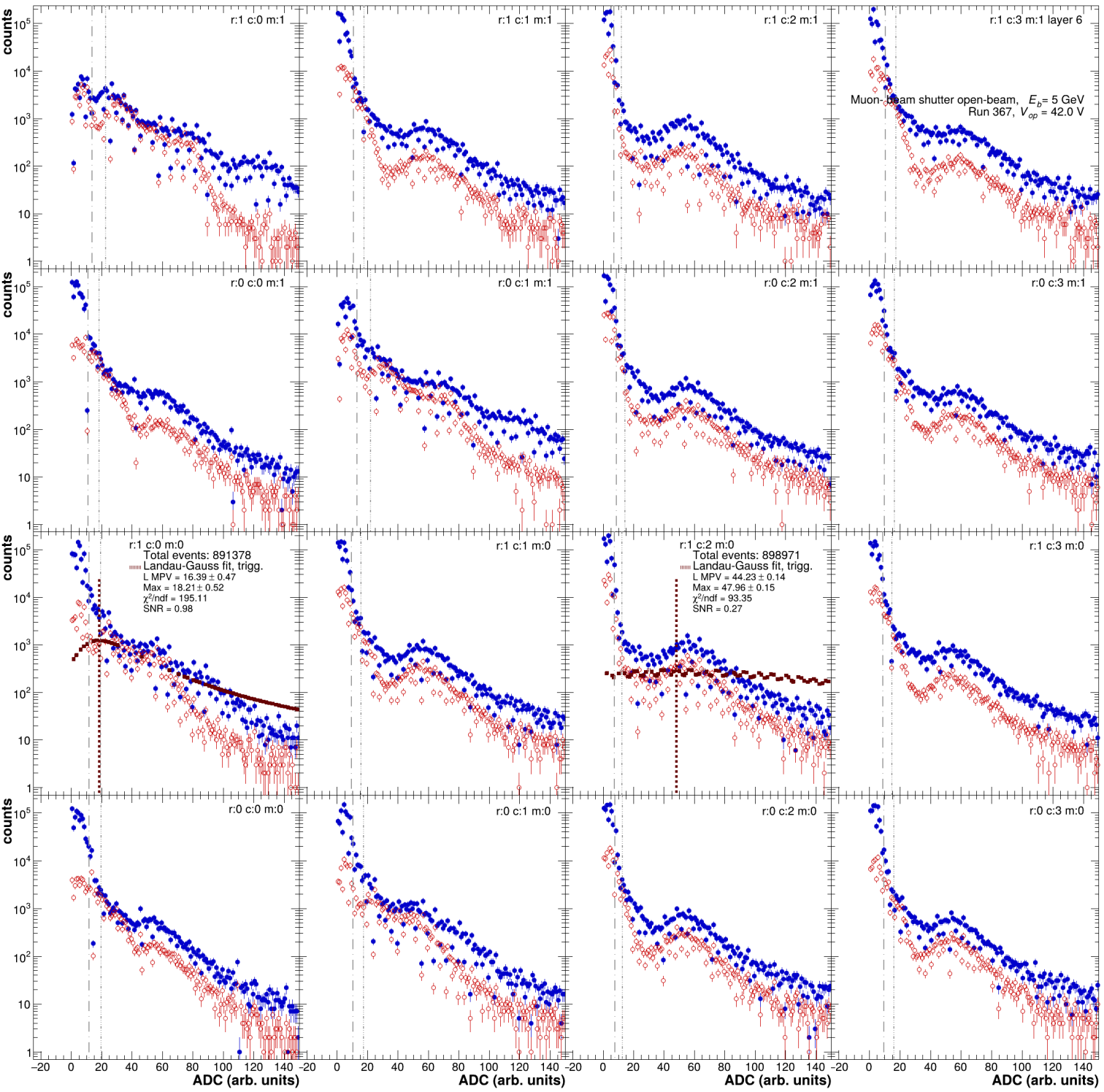

r:1 c:0 m:1
Total events: 643089
Landau-Gauss fit, trigg.
L MPV = 13.56 ± 0.11
Max = 17.21 ± 0.14
χ²/ndf = 340.31
SNR = 2.74
r:1 c:1 m:1
Total events: 858318
Landau-Gauss fit, trigg.
L MPV = 5.11 ± 0.14
Max = 5.39 ± 0.15
χ²/ndf = 363.99
SNR = 2.33
r:1 c:2 m:1
Total events: 851089
Landau-Gauss fit, trigg.
L MPV = 2.37 ± 0.32
Max = 7.51 ± 1.02
χ²/ndf = 394.35
SNR = 2.28
r:1 c:3 m:1 layer 1
Muon- beam, shutter open-beam, E_b = 5 GeV
Run 367, V_{op} = 42.0 V
r:0 c:0 m:1
Total events: 639541
Landau-Gauss fit, trigg.
L MPV = 6.78 ± 0.27
Max = 10.60 ± 0.42
χ²/ndf = 408.21
SNR = 2.16
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 899039
Landau-Gauss fit, trigg.
L MPV = 8.87 ± 0.21
Max = 12.89 ± 0.31
χ²/ndf = 221.16
SNR = 2.19
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

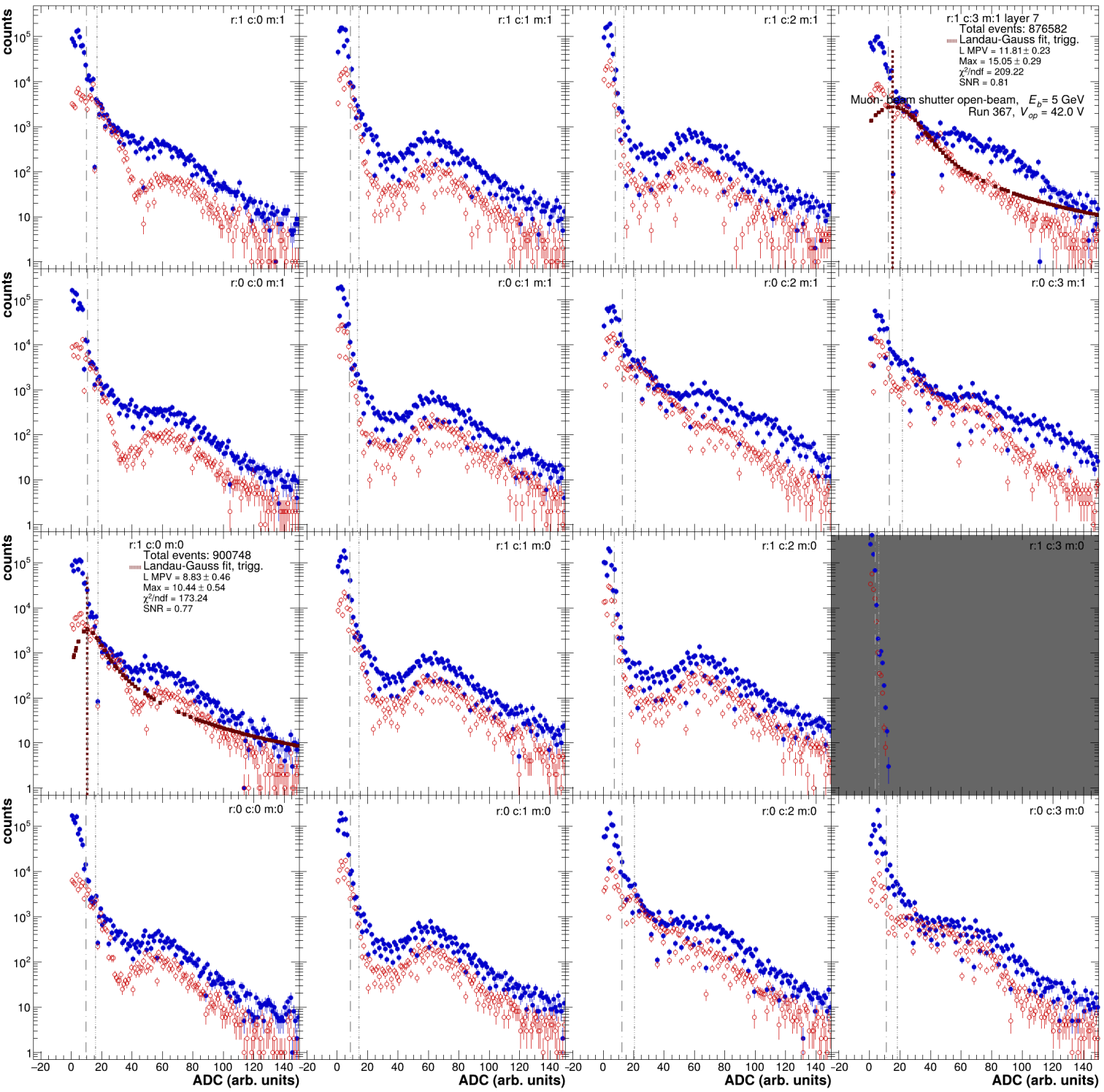

r:1 c:0 m:1
Total events: 620173
Landau-Gauss fit, trigg.
L MPV = 11.97 ± 0.14
Max = 15.20 ± 0.17
χ²/ndf = 382.50
SNR = 3.17
r:1 c:1 m:1
Total events: 875092
Landau-Gauss fit, trigg.
L MPV = 5.40 ± 0.03
Max = 6.25 ± 0.04
χ²/ndf = 249.39
SNR = 1.23
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
Muon- beam shutter open-beam, E_b = 5 GeV
Run 367, V_op = 42.0 V
r:0 c:0 m:1
Total events: 646399
Landau-Gauss fit, trigg.
L MPV = 7.59 ± 0.26
Max = 10.59 ± 0.37
χ²/ndf = 351.09
SNR = 2.24
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
Total events: 897782
Landau-Gauss fit, trigg.
L MPV = 6.57 ± 0.18
Max = 10.45 ± 0.29
χ²/ndf = 278.05
SNR = 2.81
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

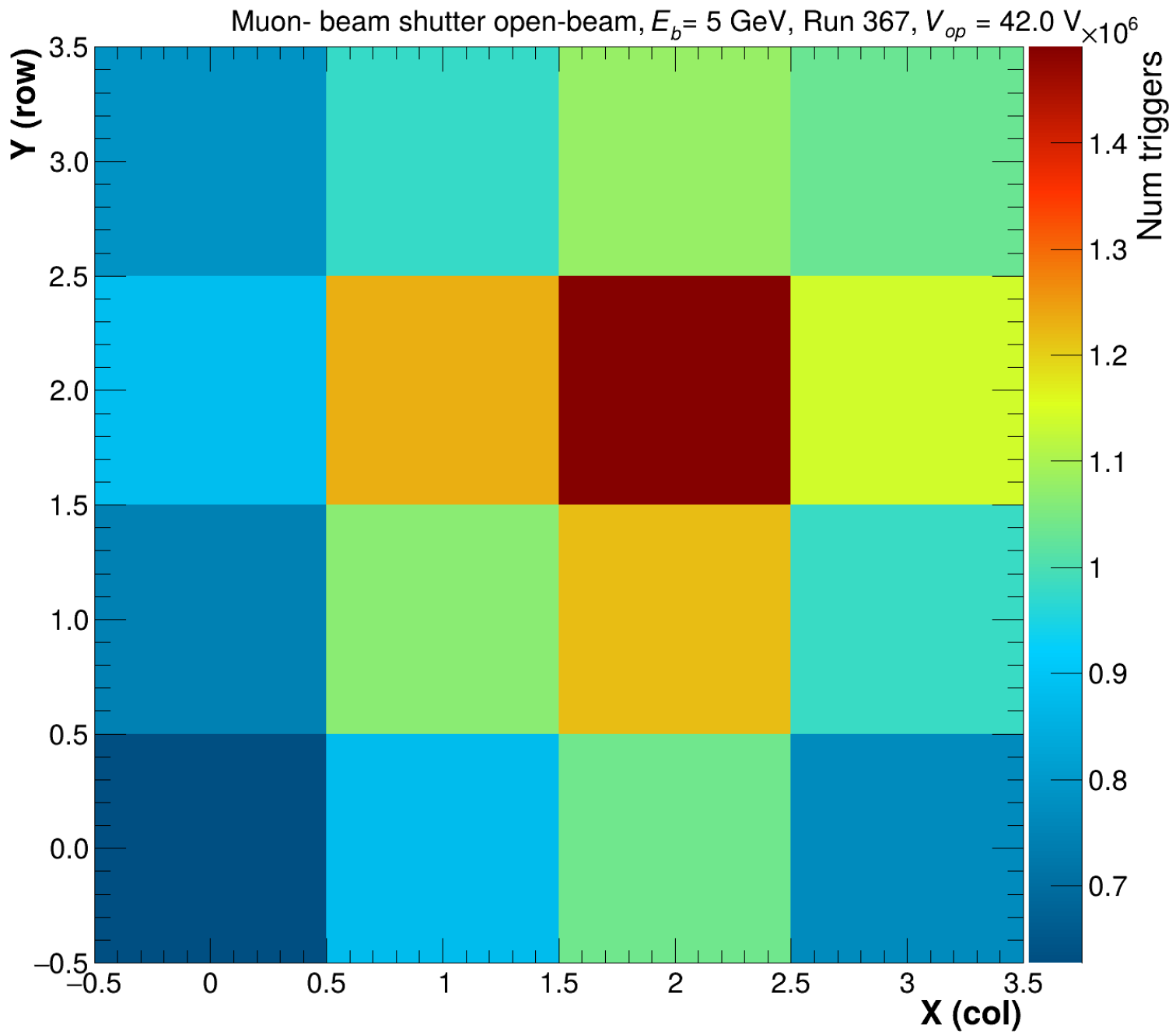

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
Total events: 880610
Landau-Gauss fit, trigg.
L MPV = 5.71 ± 0.04
Max = 7.31 ± 0.05
χ²/ndf = 327.29
SNR = 1.50
Muon: beam shutter open-beam, E_b = 5 GeV
Run 367, V_op = 42.0 V
r:0 c:0 m:1
Total events: 886171
Landau-Gauss fit, trigg.
L MPV = 16.24 ± 0.00
Max = 15.58 ± 0.00
χ²/ndf = 1570.04
SNR = 0.45
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

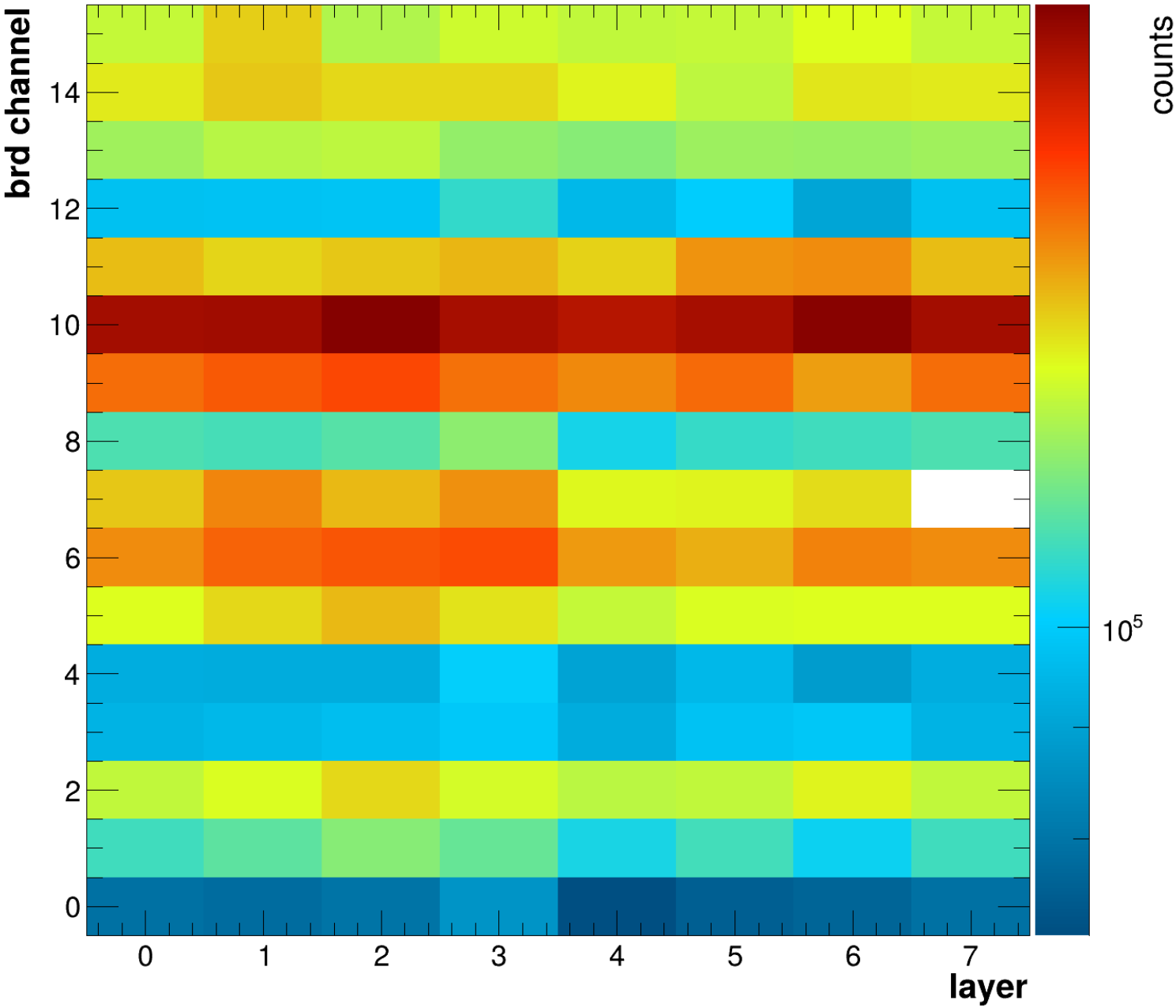

r:1 c:0 m:1
Total events: 889247
Landau-Gauss fit, trigg.
L MPV = 14.11 ± 0.07
Max = 15.23 ± 0.07
χ²/ndf = 252.29
SNR = 1.89
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
Muon- beam shutter open-beam, E_b = 5 GeV
Run 367, V_{op} = 42.0 V
r:0 c:0 m:1
r:0 c:1 m:1
Total events: 885774
Landau-Gauss fit, trigg.
L MPV = 9.88 ± 0.50
Max = 10.63 ± 0.54
χ²/ndf = 261.43
SNR = 1.02
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

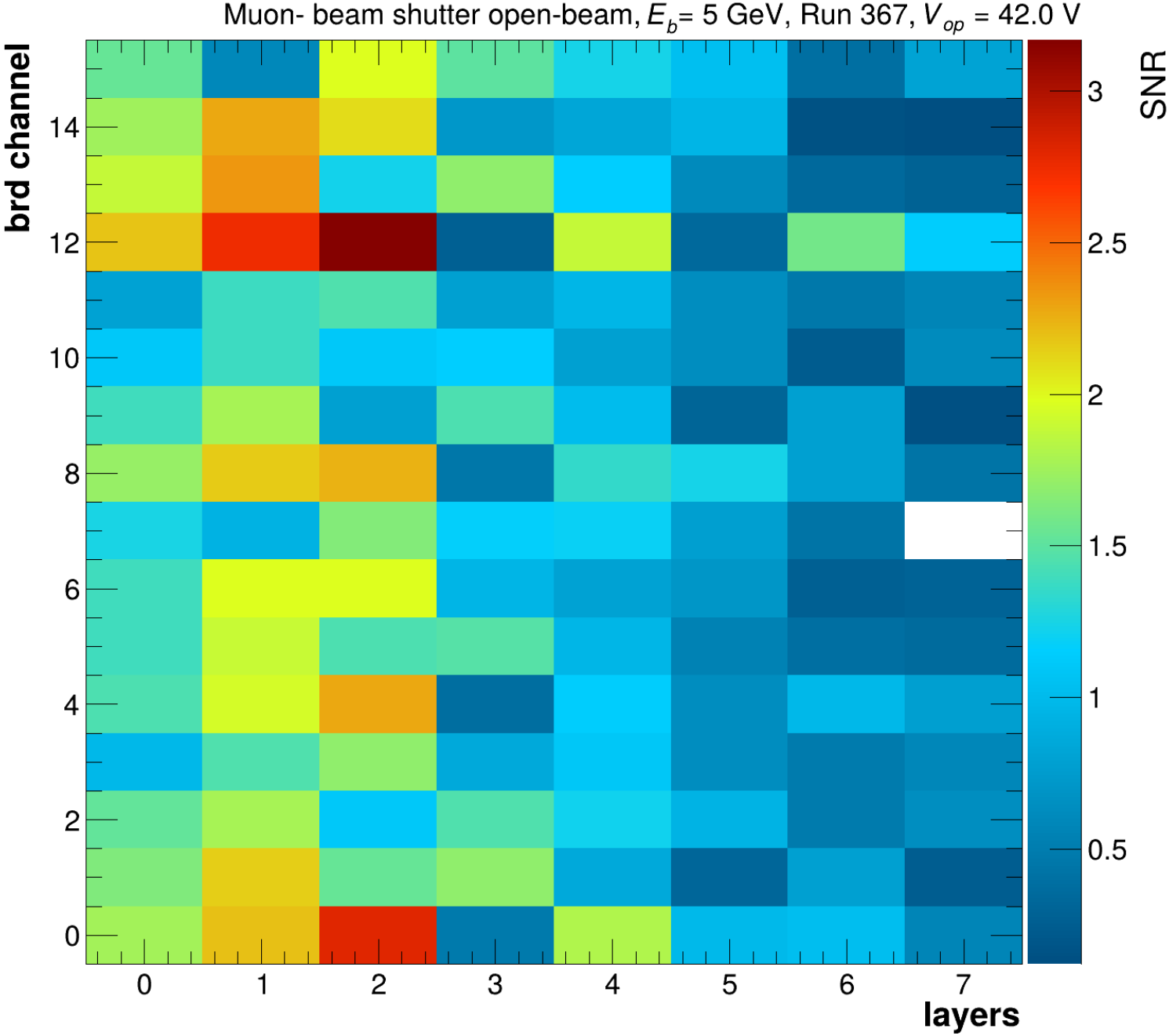

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
Total events: 886238
Landau-Gauss fit, trigg.
L MPV = 8.46 ± 0.48
Max = 12.47 ± 0.71
χ²/ndf = 245.62
SNR = 1.03
Muon beam, shutter open-beam, E_b = 5 GeV
Run 367, V_op = 42.0 V
r:0 c:0 m:1
Total events: 889790
Landau-Gauss fit, trigg.
L MPV = 14.65 ± 0.06
Max = 15.30 ± 0.07
χ²/ndf = 153.17
SNR = 1.23
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

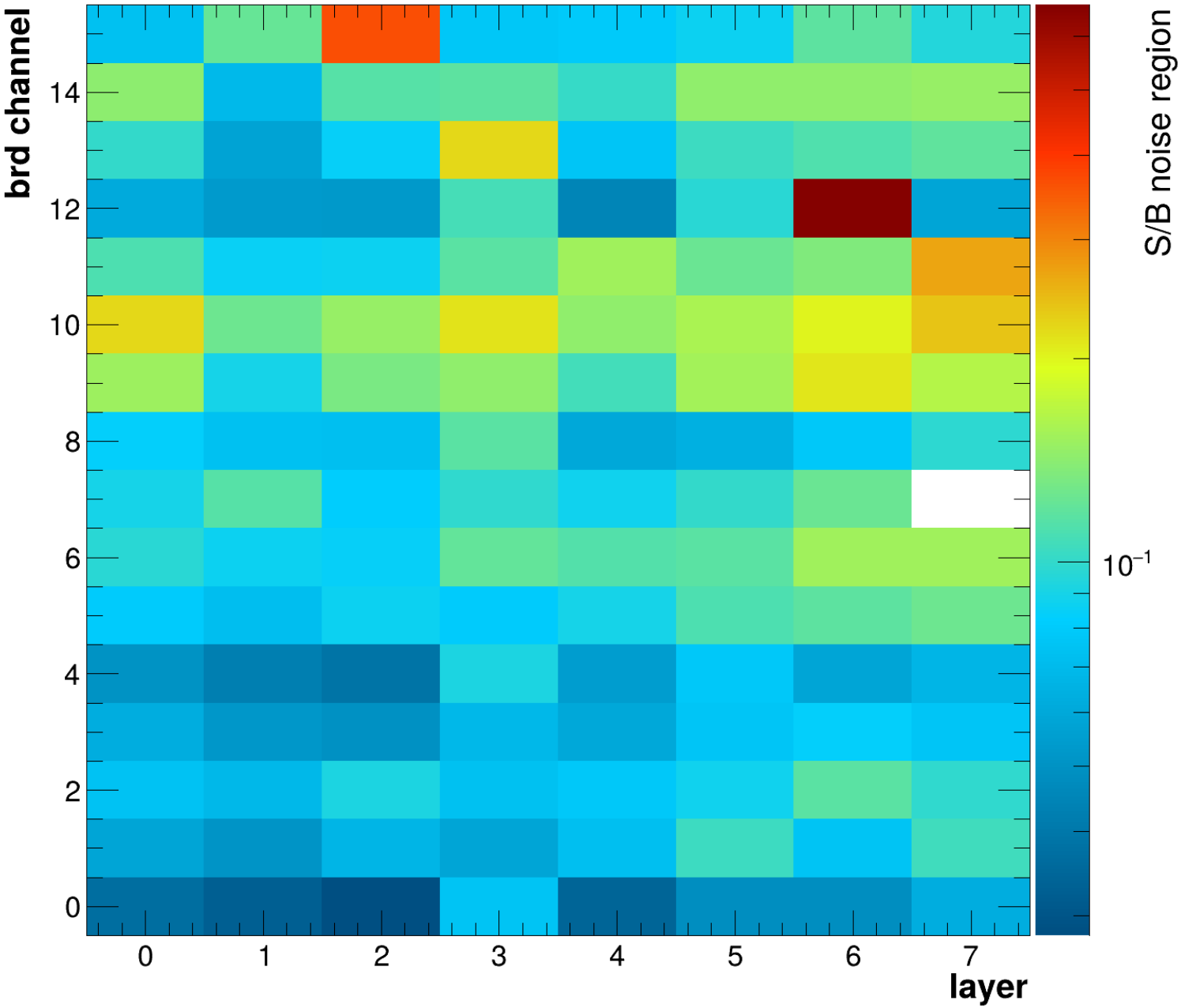

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
Muon-beam shutter open-beam, E_b = 5 GeV
Run 367, V_{op} = 42.0 V
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 891378
Landau-Gauss fit, trigg.
L MPV = 16.39 ± 0.47
Max = 18.21 ± 0.52
χ^2/ndf = 195.11
SNR = 0.98
r:1 c:1 m:0
r:1 c:2 m:0
Total events: 898971
Landau-Gauss fit, trigg.
L MPV = 44.23 ± 0.14
Max = 47.96 ± 0.15
χ^2/ndf = 93.35
SNR = 0.27
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

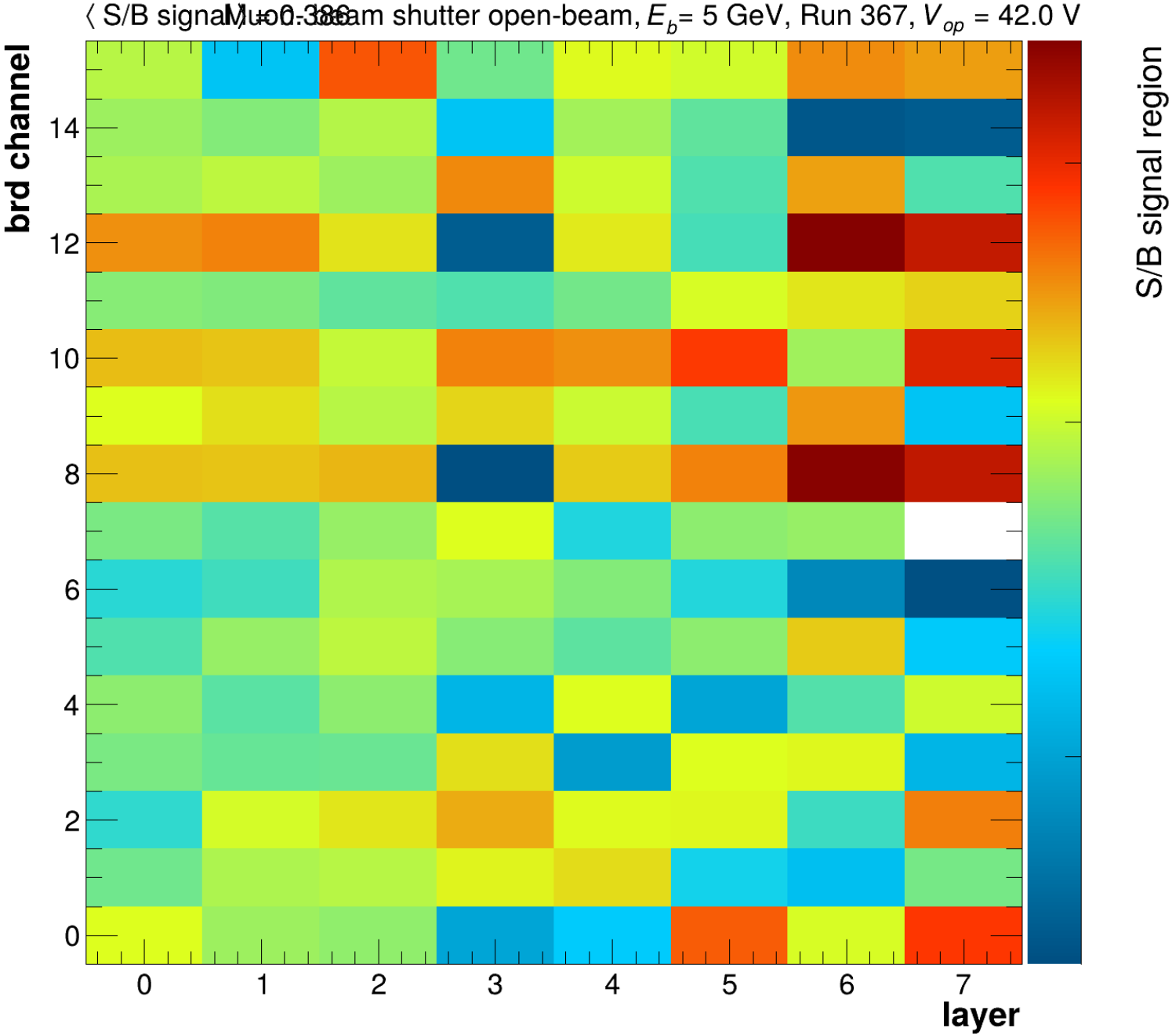

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
Total events: 876582
Landau-Gauss fit, trigg.
L MPV = 11.81 ± 0.23
Max = 15.05 ± 0.29
χ²/ndf = 209.22
SNR = 0.81
Muon-beam shutter open-beam, E_b = 5 GeV
Run 367, V_op = 42.0 V
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
Total events: 900748
Landau-Gauss fit, trigg.
L MPV = 8.83 ± 0.46
Max = 10.44 ± 0.54
χ²/ndf = 173.24
SNR = 0.77
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

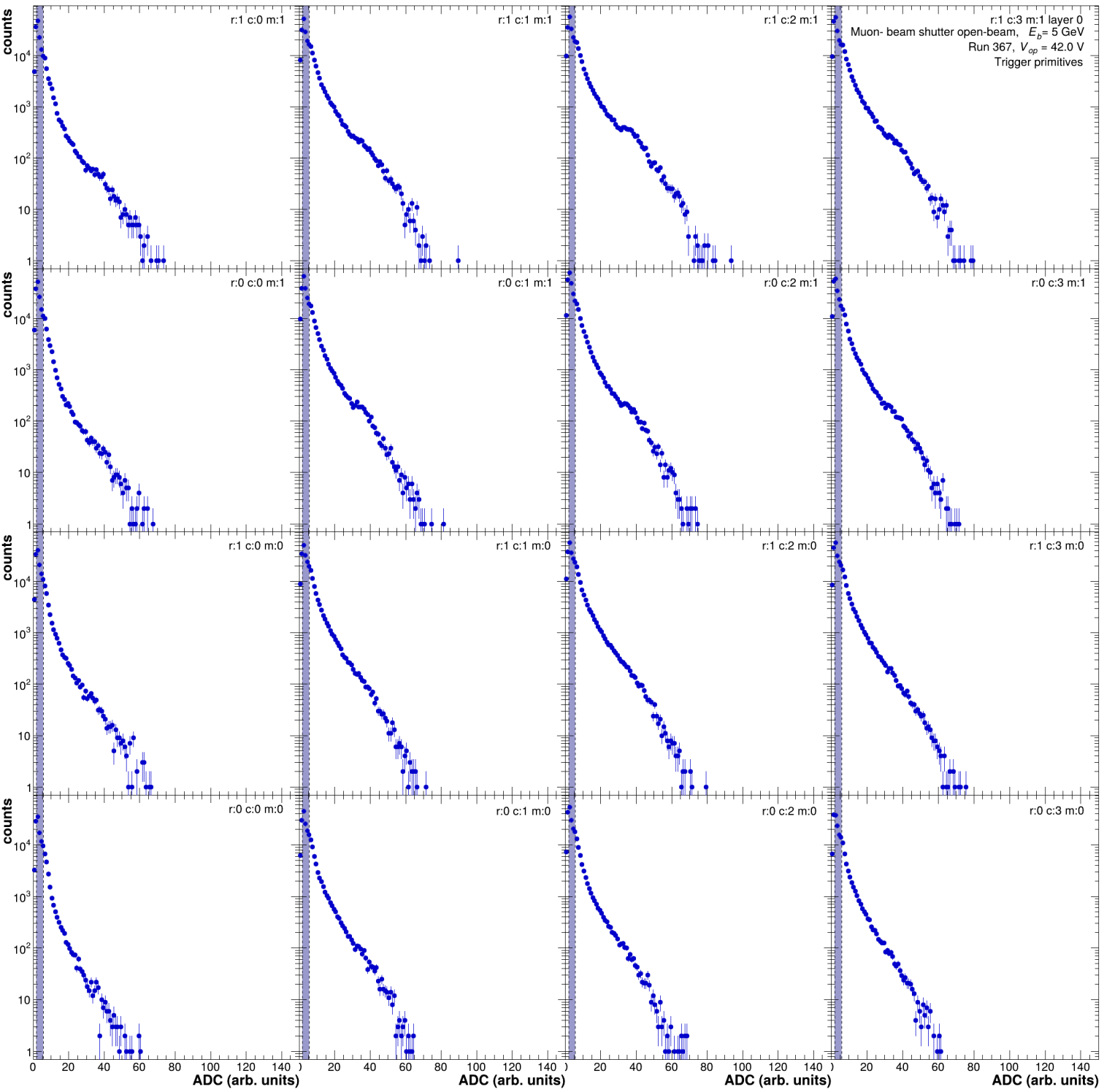

Muon- beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
Y (row)
X (col)
Num triggers
×10^6
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−0.5
0
0.5
1
1.5
2
2.5
3
3.5
1.4
1.3
1.2
1.1
1
0.9
0.8
0.7

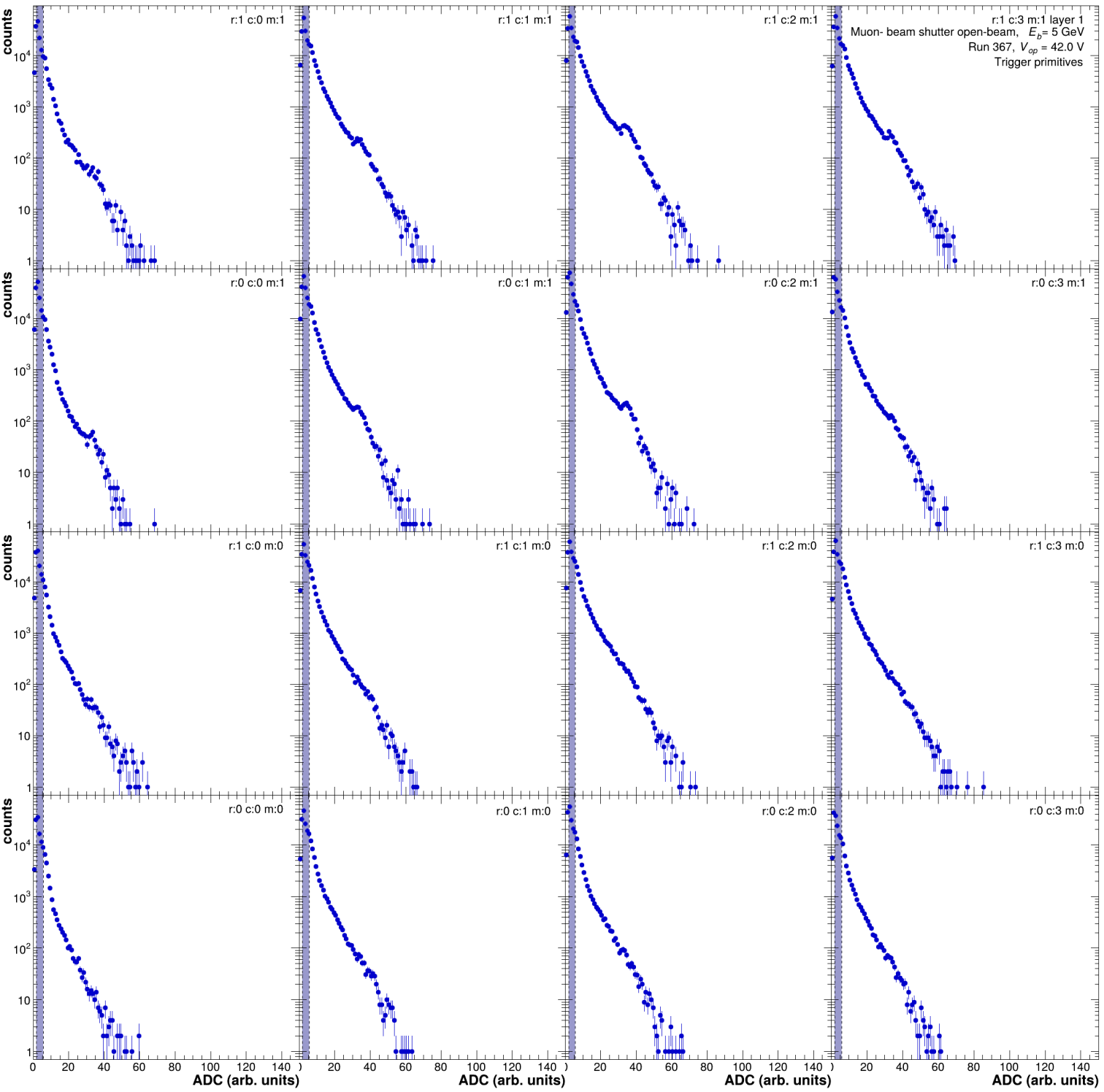

evt. collected in beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
layer
counts
10^5
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

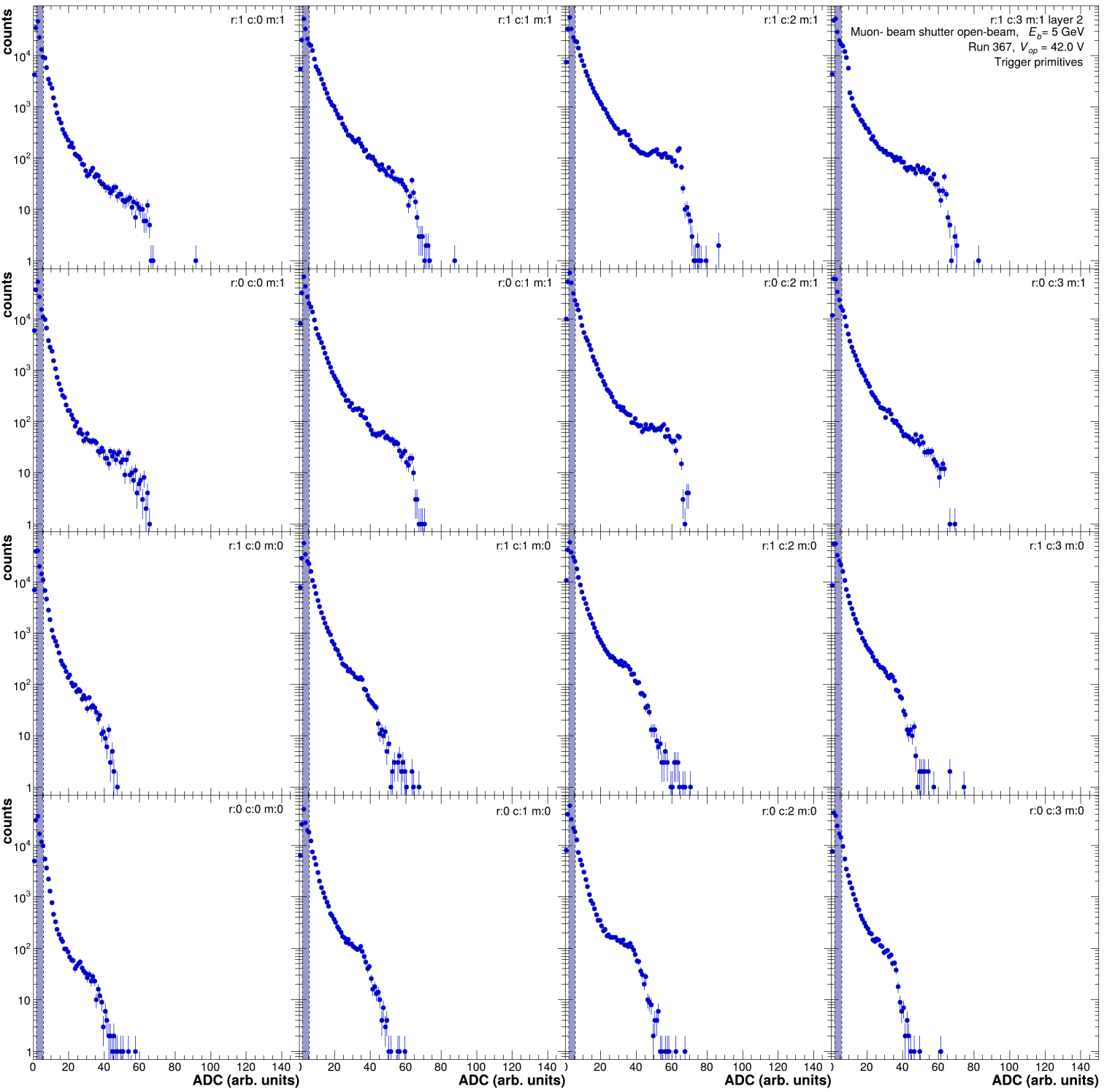

Muon- beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layers
SNR
0.5
1
1.5
2
2.5
3

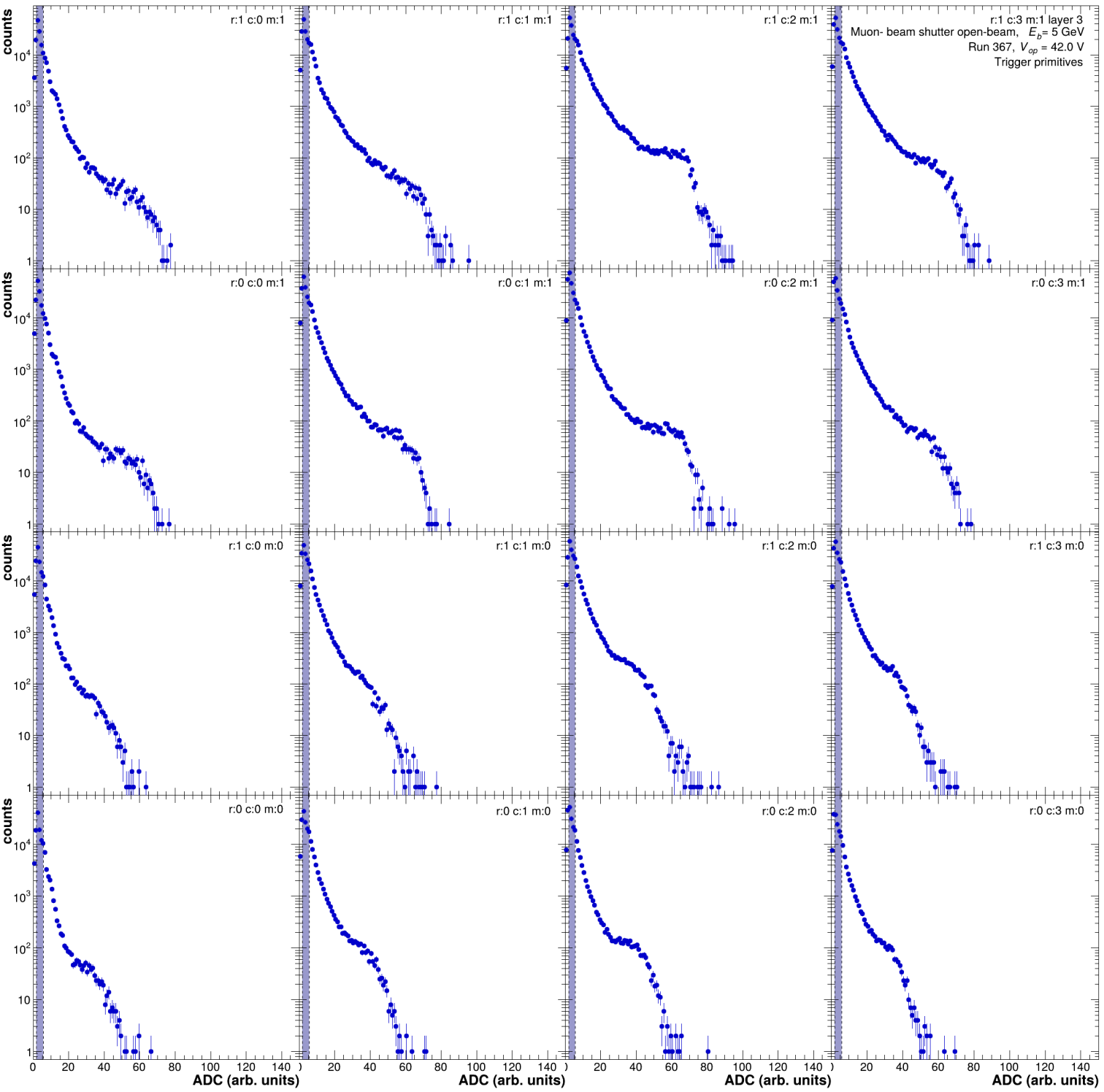

⟨ S/B noise⟩ = 0.154 beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
layer
S/B noise region
10^{-1}
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7

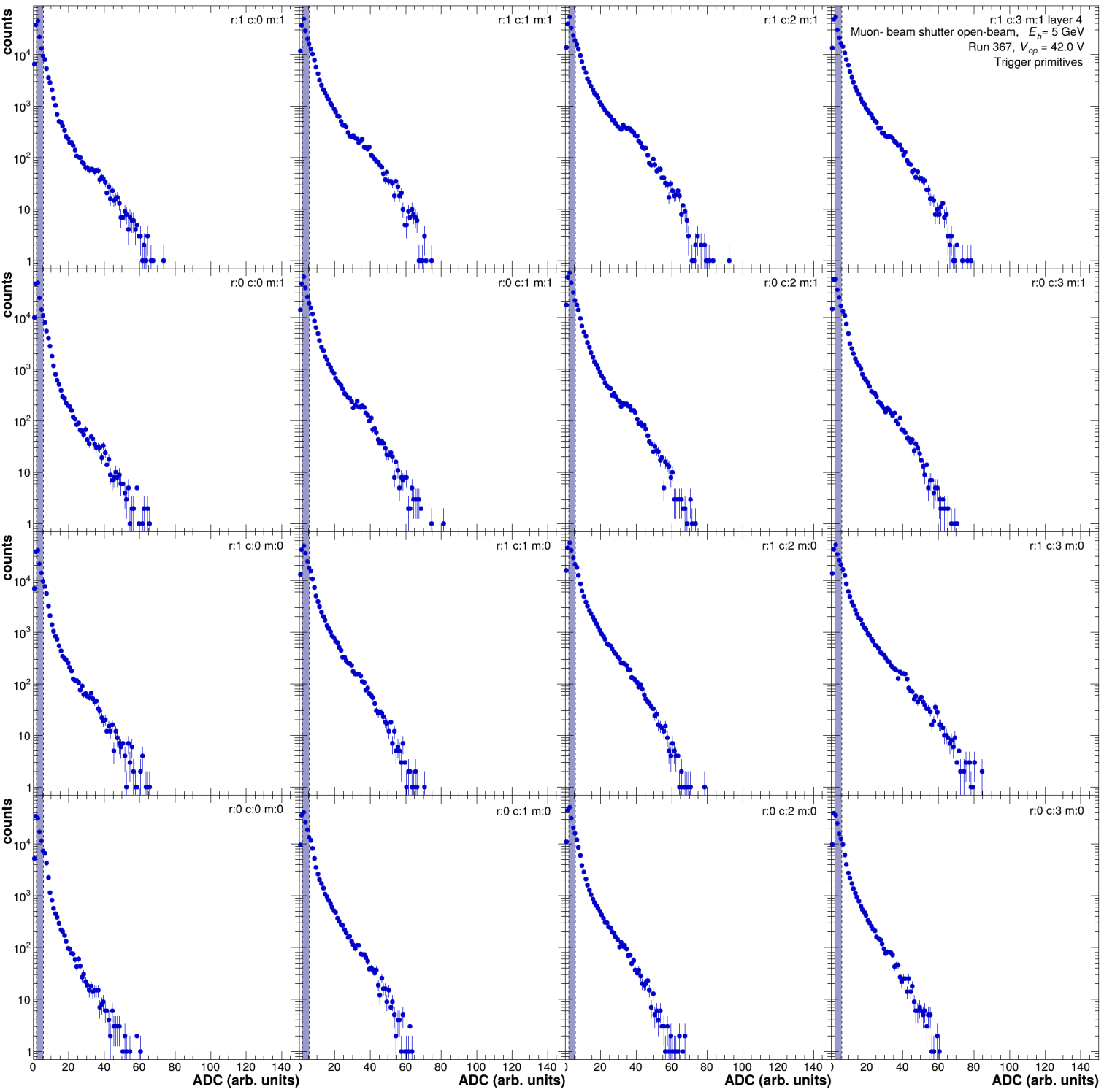

⟨ S/B signal⟩ = 0.386
Muon beam shutter open-beam, E_b= 5 GeV, Run 367, V_{op} = 42.0 V
brd channel
0
2
4
6
8
10
12
14
0
1
2
3
4
5
6
7
layer
S/B signal region

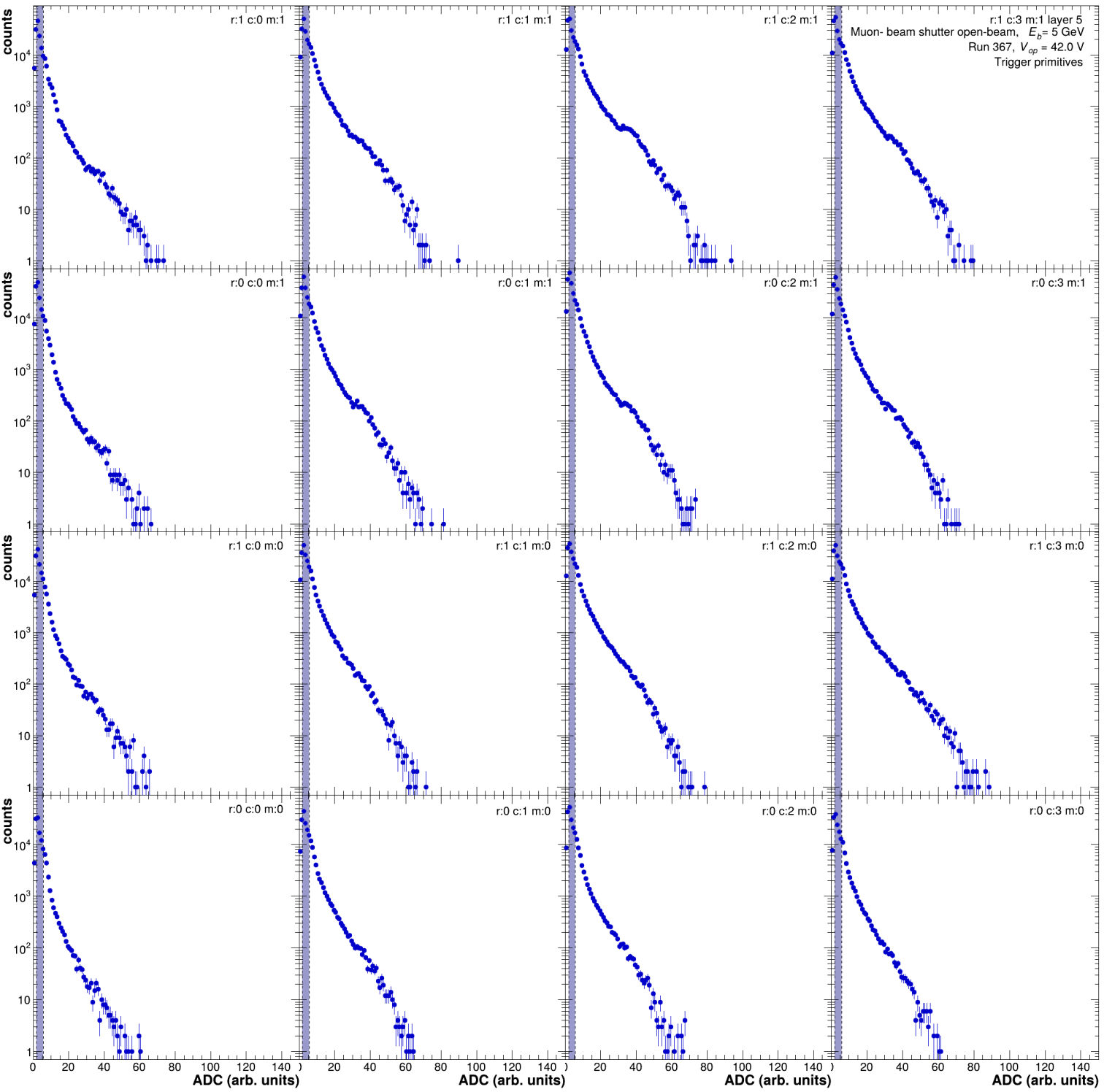

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 0
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

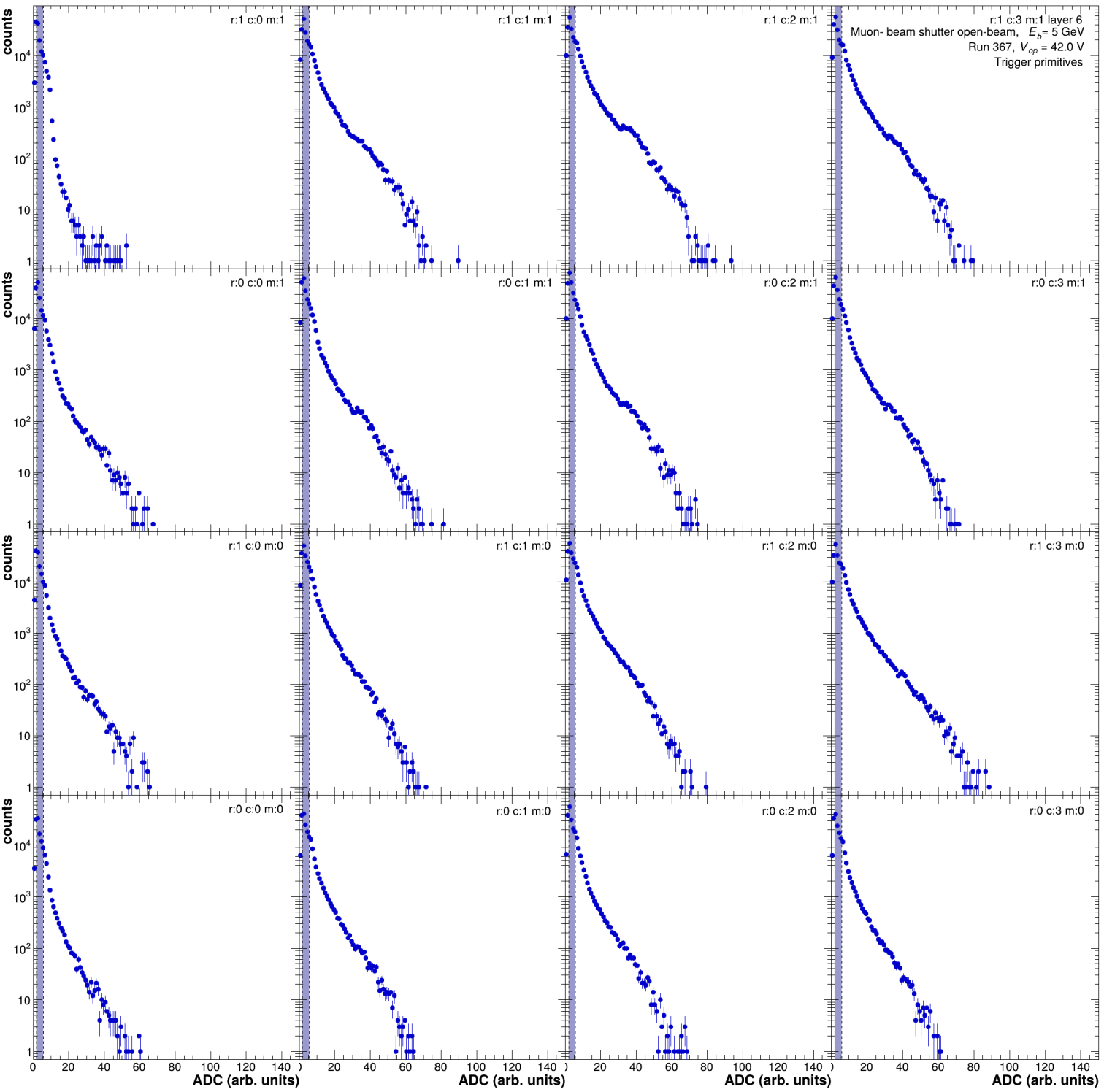

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 1
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)

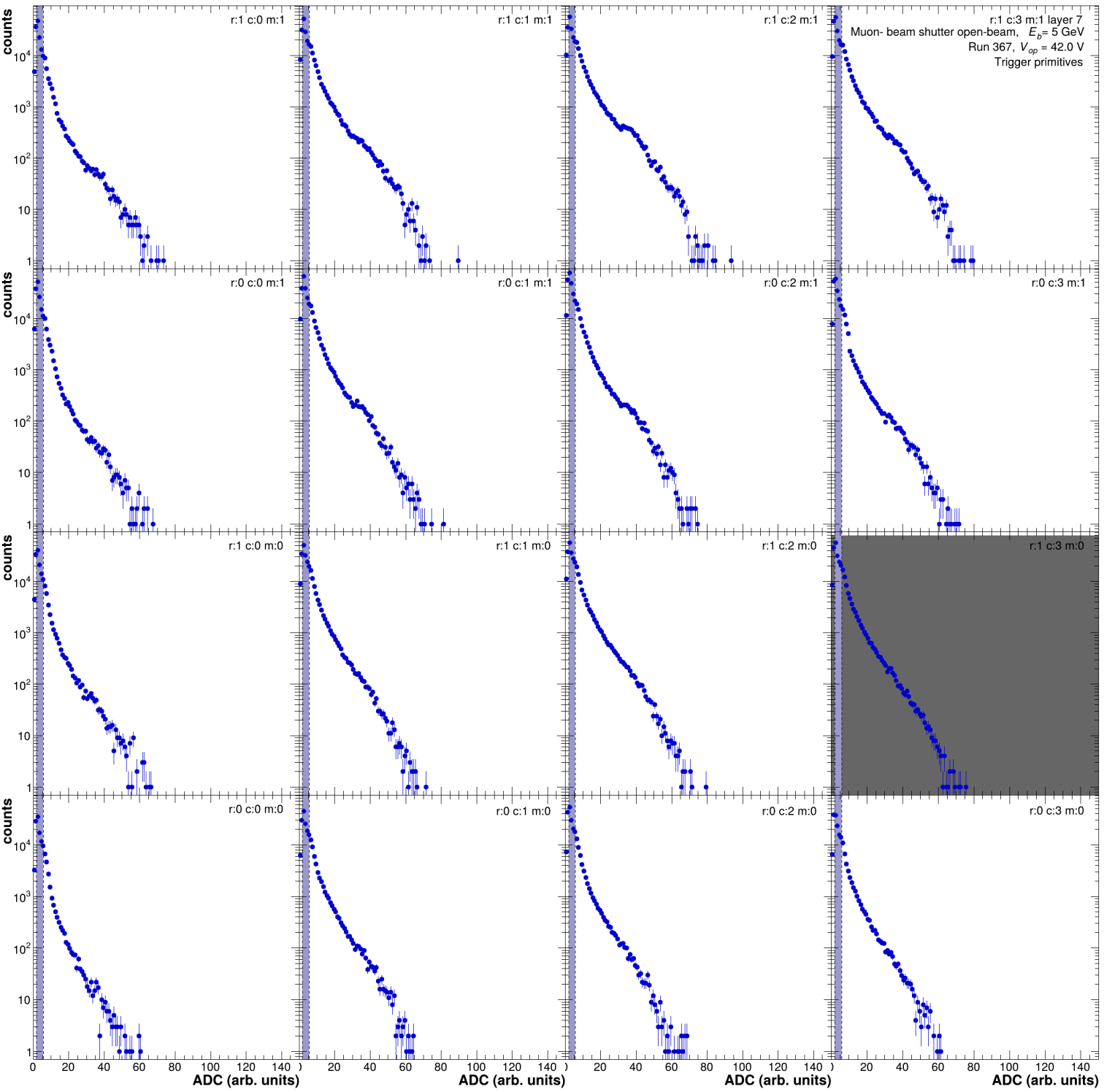

r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 2
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 3
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 4
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 5
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 6
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)


r:1 c:0 m:1
r:1 c:1 m:1
r:1 c:2 m:1
r:1 c:3 m:1 layer 7
Muon- beam shutter open-beam, E_b= 5 GeV
Run 367, V_{op} = 42.0 V
Trigger primitives
r:0 c:0 m:1
r:0 c:1 m:1
r:0 c:2 m:1
r:0 c:3 m:1
r:1 c:0 m:0
r:1 c:1 m:0
r:1 c:2 m:0
r:1 c:3 m:0
r:0 c:0 m:0
r:0 c:1 m:0
r:0 c:2 m:0
r:0 c:3 m:0
counts
ADC (arb. units)